

Exponents, Radicals & Logarithms

Strategic representation with domain and reversibility control

DOMAIN → NORMALIZE → REPRESENT → TRANSFORM → CHECK

Student learning pack · assimilation, recognition, transfer, mastery

Exponents, Radicals and Logarithms: Protect the Domain Before You Transform

Many exponent, radical and logarithm problems look as if they require a clever trick. Usually the real difficulty comes earlier: choosing a representation that keeps the equation equivalent and remembering which values are legal.

The central habit is:

Domain first. Normalize next. Transform only when you know whether the move is reversible.

A useful working loop is:

DOMAIN → NORMALIZE → REPRESENT → TRANSFORM → CHECK

1. Exponents are a representation choice

Expressions such as

$$8^x, 4^{x+1}, \text{ and } 2^{3x}$$

may look different, but they all use the same base after normalization:

$$8^x = 2^{3x},$$

$$4^{x+1} = 2^{2x+2}.$$

If an equation can be written with one positive base different from 1, equal powers give equal exponents.

Example

Solve

$$8^{x-1} = 4^{x+2}.$$

Write both sides in base 2:

$$2^{3x-3} = 2^{2x+4}.$$

So

$$3x - 3 = 2x + 4,$$

hence $x = 7$.

A logarithm would work, but it is more machinery than the problem needs.

2. Exponent rules have conditions

For nonzero a ,

$$a^m a^n = a^{m+n}$$

and

$$a^m / a^n = a^{m-n}.$$

Also

$$a^{-n} = 1/a^n,$$

so a negative exponent silently introduces the condition $a \neq 0$.

For rational exponents, it is safest in this course to normalize with a positive base unless the expression is already known to be real. Do not use a familiar exponent law in a domain where the expression itself may be undefined.

3. Principal square roots are non-negative

The symbol $\sqrt{u}$ means the non-negative square root of u , so over the real numbers:

- $u \geq 0$ is required;
- $\sqrt{u} \geq 0$;
- $\sqrt{x^2} = |x|$, not automatically x .

This sign convention controls almost every radical equation.

Example

$$\sqrt{(x-3)^2} = 5$$

means

$$|x-3| = 5,$$

so

$$x = 8 \text{ or } x = -2.$$

Writing $\sqrt{(x-3)^2} = x-3$ would lose one solution and would be false whenever $x < 3$.

4. Simplify radicals only when the domain permits it

If $u, v \geq 0$, then

$$\sqrt{uv} = \sqrt{u}\sqrt{v}.$$

For example,

$$\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}.$$

But do not split a square root across subtraction:

$$\sqrt{u-v} \text{ is not } \sqrt{u} - \sqrt{v} \text{ in general.}$$

For instance,

$$\sqrt{9-4} = \sqrt{5},$$

while

$$\sqrt{9} - \sqrt{4} = 1.$$

This distinction becomes crucial in nested radicals.

5. Conjugates remove a difference of squares

When a denominator or expression contains

$$a + b\sqrt{c}$$

or

$$\sqrt{p} + \sqrt{q},$$

the conjugate changes the mixed radical expression into a difference of squares.

Example

Simplify

$$1/(\sqrt{5} - \sqrt{2}).$$

Multiply numerator and denominator by $\sqrt{5} + \sqrt{2}$:

$$(\sqrt{5} + \sqrt{2})/(5 - 2)$$

$$= (\sqrt{5} + \sqrt{2})/3.$$

The conjugate is useful because the product has a predictable structure. It is not a ritual to apply to every radical.

6. Nested radicals: name the inside before squaring

A nested radical such as

$$\sqrt{x - \sqrt{x+a}}$$

contains two layers of domain information:

$$x + a \geq 0$$

and

$$x - \sqrt{x+a} \geq 0.$$

Squaring immediately can hide these conditions.

A better first question is:

What simple quantity can represent the inner radical or the expression left after one justified square?

A structural example

Suppose

$$\sqrt{5 + 2\sqrt{6}}$$

has the form

$$\sqrt{m} + \sqrt{n}.$$

Squaring the proposed form gives

$$m + n + 2\sqrt{mn}.$$

So we want

$$m + n = 5, mn = 6.$$

Thus $\{m, n\} = \{2, 3\}$ and

$$\sqrt{5 + 2\sqrt{6}} = \sqrt{2} + \sqrt{3}.$$

We matched structure instead of repeatedly squaring an equation.

7. Squaring: equivalence or implication?

If

$$A = B,$$

then certainly

$$A^2 = B^2.$$

But the reverse is not always true: $A^2 = B^2$ allows $A = B$ or $A = -B$.

So in general,

$$A = B \Rightarrow A^2 = B^2.$$

Squaring becomes reversible when both sides are already known to be non-negative:

$$A \geq 0, B \geq 0 \text{ and } A = B$$

is equivalent to

$$A^2 = B^2.$$

Example: an implication-only square

Solve

$$\sqrt{x+2} = x.$$

The left side is non-negative, so any solution must satisfy $x \geq 0$ and $x+2 \geq 0$.

Now both sides are non-negative, so squaring is reversible on this restricted domain:

$$x+2 = x^2.$$

Thus

$$x = 2 \text{ or } x = -1.$$

The domain condition $x \geq 0$ rejects -1 , leaving $x = 2$.

The original equation is the final authority.

8. Do not cancel a sign condition

Suppose an equation is rearranged to

$$\sqrt{U} = V.$$

Before squaring, write

$$U \geq 0 \text{ and } V \geq 0.$$

After squaring, solve

$$U = V^2$$

under those conditions.

This one habit prevents most extraneous-root errors.

9. Logarithms are exponents written backwards

For

$$a > 0, a \neq 1, b > 0,$$

$$\log_a b = t$$

means exactly

$$a^t = b.$$

The logarithm is not a new kind of arithmetic object; it records an exponent.

Example

$$\log_2 32 = 5$$

because

$$2^5 = 32.$$

When a logarithm equation becomes simpler as an exponent relation, convert it.

10. The logarithm domain is part of the equation

For $\log_a b$ over the reals:

- base $a > 0$;
- base $a \neq 1$;
- argument $b > 0$.

These conditions must be written before algebraic manipulation if variables occur in the base or argument.

Example

Solve

$$\log_{x-1} 8 = 3.$$

The base conditions require

$$x - 1 > 0, x - 1 \neq 1.$$

Convert to exponent form:

$$(x - 1)^3 = 8.$$

$$\text{So } x - 1 = 2, \text{ hence } x = 3.$$

But then the base is 2, which is legal. Therefore $x = 3$ is valid.

11. Reciprocal logarithms

When both logarithms are defined and nonzero,

$$\log_a b \cdot \log_b a = 1.$$

A clean way to see this is to let

$$t = \log_a b.$$

Then

$$b = a^t,$$

so

$$a = b^{1/t},$$

and therefore

$$\log_b a = 1/t.$$

This substitution is often cheaper than applying several memorized logarithm identities.

12. Historical pattern: convert logs to an exponent relation

In IOQM-2023-Q02, set

$$t = \log_a b.$$

Because the source has $a, b \geq 2$, both logs are legal and $t > 0$. The relation becomes

$$t + 6/t = 5,$$

so

$$t = 2 \text{ or } t = 3.$$

Therefore

$$b = a^2 \text{ or } b = a^3.$$

The rest is an integer bound count. The logarithm disappears early because exponent form is the cheaper representation.

The lesson is:

When two reciprocal logarithms appear, compress them to one exponent variable.

13. Historical pattern: a nested radical is not a difference of radicals

In IOQM-2025-Q28, the controlled equation contains

$$\sqrt{x - \sqrt{x + a}}.$$

The inner radical is part of the radicand of the outer radical. It must not be flattened into

$$\sqrt{x} - \sqrt{x + a}.$$

The successful route is:

1. record the principal-root signs;
2. prove the integer y must be 0;
3. square only after both sides are known non-negative;
4. reach $\sqrt{x + a} = x - a$;
5. set the non-negative integer $t = x - a$;
6. obtain $a = t(t - 1)/2$;
7. apply the bound and nonsquare condition.

The largest admissible value is 91.

The lesson is:

Preserve the nesting, use the sign information, and turn the final radical into an integer parameter.

14. Common base or logarithm?

Use a common base when both sides naturally become powers of the same positive base.

Use logarithms when: - the variable is in an exponent and no convenient common base exists; - a logarithm relation is already present; - exponent comparison becomes simpler after taking logs and all positivity conditions are secure.

Do not introduce logarithms merely because an exponent appears.

15. A compact router

When you meet an exponent, radical or logarithm problem, ask:

1. What values are legal?
2. Is there a principal-root sign condition?
3. Can I normalize to a common base?
4. Is a conjugate exposing a difference of squares?
5. Is the radical nested? What should I name before squaring?
6. If I square, is the move $\Leftrightarrow$ or only $\Rightarrow$?
7. Would exponent form be cheaper than logarithm form, or vice versa?
8. Are there integer restrictions to apply only after the main structure is solved?
9. Have I checked every candidate in the original condition?

The goal is not to perform more algebra. It is to keep every transformation legal and purposeful.

First-Step Reference: Exponents, Radicals and Logarithms

Use the smallest valid first move. Do not start by expanding or squaring unless the structure requires it.

You see	First question	First mathematical line
several powers such as $8^x, 4^{x+1}$	can they share one base?	rewrite them as powers of the same positive base
negative exponent	can the base be zero?	record base $\neq 0$, then use $a^{-n} = 1/a^n$
$\sqrt{U}$	is the radical defined?	write $U \geq 0$
$\sqrt{U} = V$	is the right side non-negative?	write $U \geq 0$ and $V \geq 0$ before squaring
$\sqrt{x^2}$	what sign can x have?	write $\sqrt{x^2} = x $
$\sqrt{p} - \sqrt{q}$ in a denominator	will the conjugate collapse it?	multiply by $\sqrt{p} + \sqrt{q}$
$\sqrt{A + 2\sqrt{B}}$	can it be $\sqrt{m} + \sqrt{n}$?	compare $m + n = A$ and $mn = B$
nested radical such as $\sqrt{x - \sqrt{x + a}}$	what inner quantity should be named or isolated?	preserve the nesting; record both radicand conditions
equation after a square	did the square preserve equivalence?	mark the move $\Leftrightarrow$ if both sides had known same sign; otherwise mark $\Rightarrow$ and plan a final check
$\log_a b$	is the logarithm legal?	write $a > 0, a \neq 1, b > 0$
reciprocal logs $\log_a b, \log_b a$	can one variable represent both?	set $t = \log_a b$, then $\log_b a = 1/t$
variable exponent, no common base	would log form reduce the exponent cleanly?	take logs only after positivity/domain is secure
integer restrictions after radicals/logs	has the continuous algebraic structure already been solved?	finish the equation first, then apply the integer filter

Fast route choices

Common base vs logarithm

Prefer **common base** when both sides already factor into powers of one small positive base.

Prefer **logarithm** when no useful common base exists and taking logs genuinely makes the variable exponent linear or simpler.

Conjugate vs squaring

Prefer a **conjugate** when a sum/difference of radicals is multiplying or dividing and a difference of squares will remove the radicals.

Prefer **squaring** when an equation has an isolated radical and all needed sign conditions have been recorded.

Simple radical vs nested radical

For a simple radical, isolate it and protect the sign.

For a nested radical, first preserve the inside as one object. Repeated squaring without a representation plan usually creates more structure than it removes.

The three checks before squaring

Before replacing $\sqrt{U} = V$ by $U = V^2$, confirm:

1. $U \geq 0$;
2. $V \geq 0$;
3. whether these conditions make the square reversible.

If you cannot prove the sign of V , squaring may create candidates. Check them in the original equation.

The three checks before using a logarithm

For every variable logarithm, confirm:

1. base > 0 ;
2. base $\neq 1$;
3. argument > 0 .

Then decide whether exponent form or log form is cheaper.

Stop rule

Stop transforming once the requested value is determined and every candidate satisfies the original domain and equation. More algebra is not more proof.

Recognition and First-Line Lab

For each item, write only the **best first move** or choose the best option. Do not complete a long solution unless needed to justify the choice.

Recognition

- $8^{x-1} = 4^{x+2}$. What is the cheapest representation? A. decimal approximations B. common base 2 C. take square roots D. expand powers
- Which statement is always correct for real x ? A. $\sqrt{x^2} = x$ B. $\sqrt{x^2} = -x$ C. $\sqrt{x^2} = |x|$ D. $\sqrt{x^2} = x^2$
- Before squaring $\sqrt{U} = V$, which condition is essential for a reversible square? A. $V \leq 0$ B. $U < 0$ C. $U \geq 0$ and $V \geq 0$ D. $U = V$
- Which expression most strongly suggests a conjugate as the first tool? A. $2^x = 16$ B. $1/(\sqrt{7} - \sqrt{5})$ C. $\sqrt{x+4} = 3$ D. $\log_2 8$
- What must be preserved in $\sqrt{x - \sqrt{x+a}}$? A. treat it as $\sqrt{x} - \sqrt{x+a}$ B. the inner radical as part of the outer radicand C. only the outer square root D. only the integer condition
- You square an equation without first proving both sides have the same known sign. What is safest to record? A. the square is automatically equivalent B. the square may generate candidates, so check the original C. no domain is needed D. every quadratic root is valid
- For $\log_{x-2}(x+1)$, which domain conditions belong to the first line? A. $x+1 > 0$ only B. $x-2 > 0$, $x-2 \neq 1$, $x+1 > 0$ C. $x-2 \neq 0$ only D. $x > 0$ only
- If $a, b > 1$ and both reciprocal logarithms occur, which compression is useful? A. $t = a + b$ B. $t = \log_a b$, so $\log_b a = 1/t$ C. $t = ab$ D. $t = \log_a a$

First lines

- Solve-start only: $27^x = 9^{x+1}$. Write the common-base line.
- For $\sqrt{5-2x}$, write the real-domain condition.
- For $\sqrt{2x+3} = x+1$, write the sign/domain conditions that must hold before a reversible square.
- Rationalize-start only: $1/(\sqrt{11} + \sqrt{3})$. Write the conjugate multiplier.
- For $\sqrt{13-4\sqrt{10}}$, write the two equations obtained by trying the form $\sqrt{m} - \sqrt{n}$.
- For $\sqrt{x - \sqrt{x+6}} = 2$, write the two nested-radical domain conditions before solving.
- For $\log_x 16 = 4$, write the base conditions and the exponent-form equation.
- For $\log_a b + 6 \log_b a = 5$ with $a, b \geq 2$, write the one-variable substitution and resulting equation in t .

Practice and Transfer Bank

Try each stage without looking ahead. The prompts become less explicit as you move down the page.

Reconnect

1. Simplify $27^{2/3}$.
2. Solve $9^x = 27$.
3. Simplify $\sqrt{200}$.
4. State the real values of x for which $\sqrt{5 - 2x}$ is defined.

Structure with support

5. Rationalize $1/(\sqrt{7} - \sqrt{5})$.
6. Solve $\sqrt{x + 6} = x$. Record the sign condition before squaring and check the final candidate(s).
7. Solve $\sqrt{(x - 4)^2} = 3$.
8. Write $\sqrt{7 + 4\sqrt{3}}$ in the form $\sqrt{m} + \sqrt{n}$ with positive integers m, n .

Reduced support

9. Solve $\sqrt{2x + 3} = x + 1$ over the reals.
10. Solve $\sqrt{x + 1} - \sqrt{x - 3} = 1$.
11. Simplify $\sqrt{13 - 4\sqrt{10}}$.
12. Solve $\sqrt{x - \sqrt{x + 6}} = 2$ over the reals.

Mixed independent

13. Solve $\log_2(x - 1) = 3$.
14. For natural numbers $2 \leq a, b \leq 20$, find the number of ordered pairs satisfying $\log_a b + \log_b a = 5/2$.
15. Express the real solution of $2^x = 7$ exactly using a logarithm.
16. Solve $\log_x 16 = 4$ over the reals.

Transfer

17. Historical anchor — IOQM-2023-Q02.

For natural numbers $2 \leq a, b \leq 2023$, determine the number of ordered pairs satisfying $\log_a b + 6 \log_b a = 5$.

18. Historical anchor — IOQM-2025-Q28.

Assume a is a positive integer that is not a perfect square. Non-negative integers x, y satisfy

$$\sqrt{x - \sqrt{x + a}} = \sqrt{a} - y.$$

Find the largest possible $a < 100$.

19. A student solves $\sqrt{x} = x - 2$ by squaring and reports both roots of the resulting quadratic. Explain precisely what condition must be checked and determine the valid solution(s).
20. Solve $\sqrt{\log_2 x} = 2$ over the reals, including every domain condition used.

Benchmark Assimilation Lab — Exponents, Radicals and Logarithms

Use this after the Assimilation Book and before the final mastery test. It checks whether the topic has become usable, not merely familiar.

Learning loop:

RECONNECT → DIAGNOSE → ADOPT → TRANSFER

A. RECONNECT — no notes

Write only the first useful line unless a short answer is requested.

1. Simplify $\sqrt{x^2}$ over the reals.
2. For $\sqrt{2x-1} = x-2$, what sign/domain facts must be written before squaring?
3. Which is the cheaper first route for $8^{x+1} = 4^{2x-1}$: common base or logarithms?
4. What is the conjugate of $\sqrt{7} - \sqrt{3}$, and why is it useful?
5. State the real-domain conditions for $\log_{x-1}(x+2)$.
6. In $\sqrt{x - \sqrt{x+a}}$, which error would be made by writing $\sqrt{x} - \sqrt{x+a}$?
7. If $t = \log_a b$, what is $\log_b a$ when both logs are defined?
8. If squaring an equation produces two candidates, what remains to be checked?

Diagnostic interpretation:

- misses 1-2 → principal-root/sign bridge is weak;
- misses 3 → representation choice is weak;
- misses 4 → conjugate/difference-of-squares bridge is weak;
- misses 5 or 7 → logarithm-domain/exponent bridge is weak;
- misses 6 → nested-radical representation is weak;
- misses 8 → equivalence/checking discipline is weak.

B. Error laboratory

For each proposed solution, identify the **first invalid or inefficient move** and repair it.

Error 1 — square-root sign loss

Claim: $\sqrt{(x-5)^2} = x-5$.

Repair target: state the correct expression and when the claimed simplification would become valid.

Error 2 — automatic squaring

Claim: from $\sqrt{x+3} = 1-x$, immediately square to $x+3 = (1-x)^2$ and keep every quadratic root.

Repair target: record the missing sign condition before deciding whether squaring is reversible.

Error 3 — flattening a nested radical

Claim: $\sqrt{x - \sqrt{x+a}} = \sqrt{x} - \sqrt{x+a}$.

Repair target: explain why the outer radicand must be preserved as one object.

Error 4 — logarithm before domain

Claim: solve $\log_{x-2}(x+1) = 2$ by writing $(x-2)^2 = x+1$ first.

Repair target: write the three log-domain checks first.

Error 5 — unnecessary logarithms

Claim: solve $9^x = 27$ by taking logarithms.

Repair target: choose the smaller representation.

Error 6 — integer search too early

Claim: if an equation contains integer variables and radicals, start trying integer values before reducing the radical structure.

Repair target: state the correct order of operations.

C. ADOPT — first move only

Do not solve fully. Write the first two useful lines.

1. $\sqrt{3x+4} = x$.
2. $1/(\sqrt{13} - \sqrt{5})$.
3. $\sqrt{17 + 4\sqrt{15}}$.
4. $\log_{x-1} 27 = 3$.
5. $\log_a b + 2 \log_b a = 3$, with $a, b > 1$.
6. $16^{x-1} = 8^{x+2}$.
7. $\sqrt{x} - \sqrt{x+6} = 2$.
8. A radical equation has already been squared once; one side before the second square is $x-4$. What must be checked?

D. TRANSFER — changed surface

These are not number swaps of the earlier examples.

1. A quantity u is known to satisfy $u \geq 0$ and $u^2 = 11 - 6\sqrt{2}$. Decide whether a conjugate-style square recognition is useful before taking another square root.
2. Positive integers $a, b \leq 100$ satisfy $\log_a b + \log_b a = 5/2$. Turn the logarithm statement into a finite exponent/counting problem.
3. An equation contains $\sqrt{P(x)} = Q(x)$ where $Q(x)$ changes sign. Explain why solving the squared polynomial alone cannot be the final method.
4. A nested radical equation eventually gives $\sqrt{N} = k + m\sqrt{a}$, where N, k, m, a are integers and a is nonsquare. What invariant can separate rational and irrational parts?
5. A variable occurs only in exponents, but both sides are powers of 6 after factorization. Explain why logarithms are mathematically valid but strategically inferior.
6. Construct a simple equation where squaring is genuinely reversible because both sides are already known non-negative.

E. Six-question assimilation test

Choose one problem from Section D and answer all six questions in words before completing algebra.

1. **What did you notice?** Name the visible structural clue.
2. **Why does the method work?** State the invariant or equivalence condition.
3. **What clue would make you think of it again?** Give a future recognition cue.
4. **What similar-looking situation needs a different method?** State one contrast pair.
5. **Can you write the first two useful lines without help?** Write them.
6. **Can you solve a disguised version?** Change the surface while preserving the same invariant, then outline the route.

Readiness rule

You are not ready merely because you can reproduce a worked example. You are ready when you can protect the domain, choose the smallest representation, explain why a transformation is legal, reject a tempting near-miss, and start a disguised problem without a method label.

Mixed Mastery Test

Solve each problem without method labels or hints. Show enough working to make domain restrictions and candidate checks visible.

1. Solve $16^{x-1} = 8^{2x+1}$.
2. Solve $\sqrt{3x-2} = x$ over the reals.
3. Simplify $\sqrt{(2-\sqrt{3})^2}$.
4. Rationalize $3/(\sqrt{5}+1)$.
5. Solve $\sqrt{x+5} - \sqrt{x+1} = 1$.
6. Simplify $\sqrt{9+4\sqrt{5}}$.
7. Solve $\sqrt{x-\sqrt{x+2}} = 1$ over the reals.
8. Solve $\sqrt{x} = x-2$ and justify why any rejected quadratic root is invalid.
9. Solve $\log_3(x-1) = 2$.
10. Solve $\log_x 27 = 3$ over the reals.
11. For natural numbers $2 \leq a, b \leq 100$, determine the number of ordered pairs satisfying $\log_a b + 2 \log_b a = 3$.
12. A positive integer $a < 50$ is not a perfect square. Suppose a non-negative integer x satisfies $\sqrt{x-\sqrt{x+a}} = \sqrt{a}$.
Find the largest possible a .